

The Stratum Morphological Model

Kangaroo-Based Structural Form with Robot-Guided Cutting and Fologram-Assisted Assembly



Cornell AAP
Architecture Art Planning

Researcher and Designer
Julia + Bo
Supervised by:
Lawson Spencer

The Stratum Morphological Model

Abstract

The Stratum Morphological Model explores a novel approach in digital fabrication, integrating Kangaroo-based structural formfinding with robotic cutting and Fologram-assisted assembly. This project aims to demonstrate the capabilities of robotic manufacturing processes combined with computational simulations to create layered, organic forms that can be fabricated and assembled efficiently. Guided by the physics of Kangaroo simulations, the model showcases how complex geometries can be translated into physically realizable structures using robotic technology.

Keywords: Kangaroo Simulation, Layered Morphology, Robotic Fabrication, Fologram, Digital Assembly

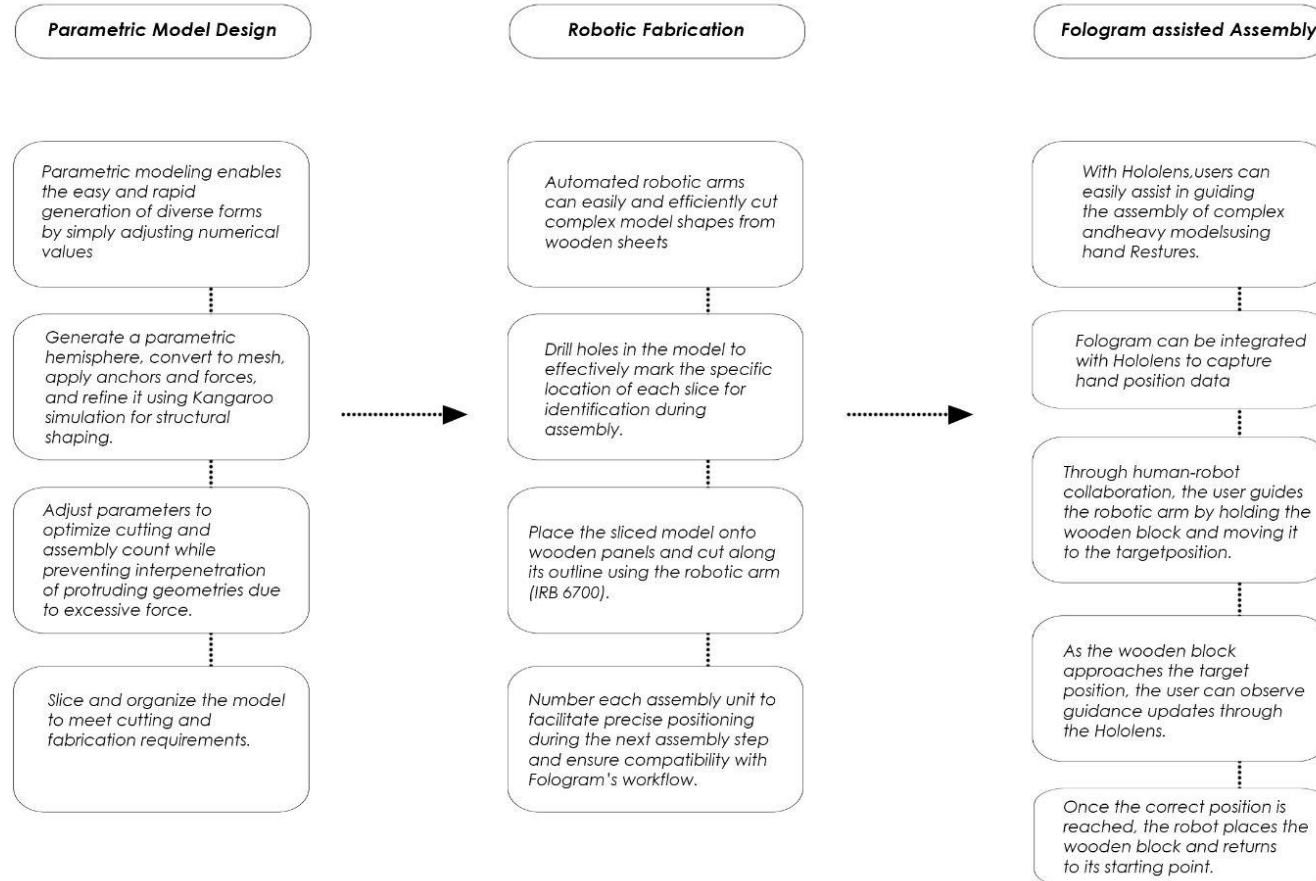


Workflow Diagram

step 01: Morphological study use Kangaroo

step 01: Morphological study use Kangaroo

step 01: Morphological study use Kangaroo



Literature Review

literature review 1:

Teleoperation of an ABB IRB 120 Robotic Manipulator and BarrettHand BH8-282 Using a Geomagic Touch X Haptic Device and ROS

<https://ieeexplore.ieee.org/document/8588549>

In the referenced project, the researchers used the Geomagic Touch X haptic device as the primary user interface, allowing the operator to control an ABB IRB 120 robotic arm equipped with a BarrettHand BH8-282 gripper. The system provided 6-DOF positional input and force feedback, enabling the user to feel resistance during teleoperation. Visual feedback was delivered through a PC screen, showing either a 3D simulation or a live camera feed of the robot's environment. This setup demonstrated the effectiveness of a haptic interface for achieving precise remote manipulation.

In contrast, our project explores the use of augmented reality (AR) to control the robot arm. Rather than operating from a first-person perspective, we asked ourselves: What advantages does AR bring to robotic fabrication? We found that AR is more suited to collaborative interaction between humans and robots. Instead of commanding the robot remotely, we aim to guide and assist the robot physically—moving and placing wooden components through a shared effort between the user's hand and the robot arm, creating a more intuitive and cooperative form of assembly.

Recently, there has been a growing number of projects using HoloLens for manual assembly, where users rely on augmented reality (AR) to receive visual guidance and assemble components by hand. Many such examples can be easily found online, demonstrating AR's usefulness in human-guided assembly. However, when it comes to AR-integrated robotic fabrication, particularly using robotic arms to assist in physical construction tasks, the number of projects is significantly smaller. Among the few that do exist, most focus on gesture-based control systems—where users perform specific actions (e.g., pointing, twisting, swiping) to command the robot. While this interaction model resembles gaming and can be intuitive for demonstrations, it often limits the potential of AR-robot collaboration. This is because users must first design a fixed set of gestures and then perform them precisely, which adds cognitive and physical load and reduces flexibility during dynamic tasks like real-time assembly.

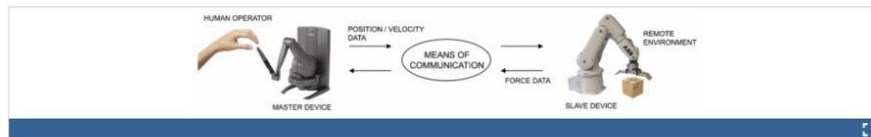


Fig. 4:
Haptic teleoperation system structure.



Fig. 2:
ABB IRB 120, barretthand bh8-282 and geomagic touch X.

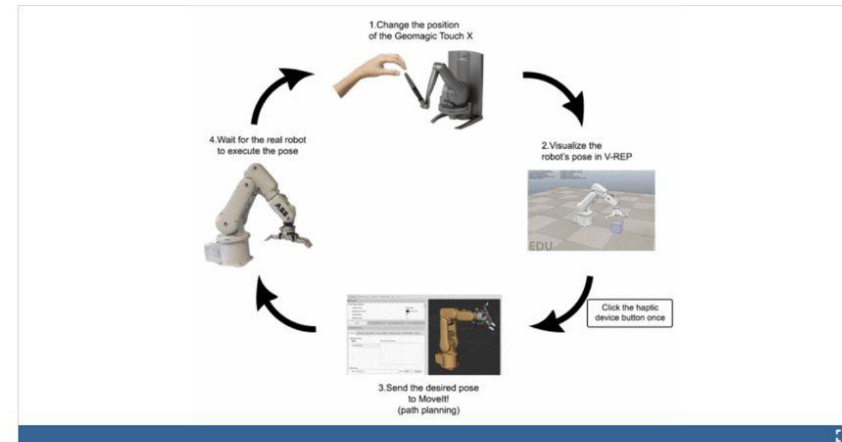


Fig. 10:
Operation flow chart.

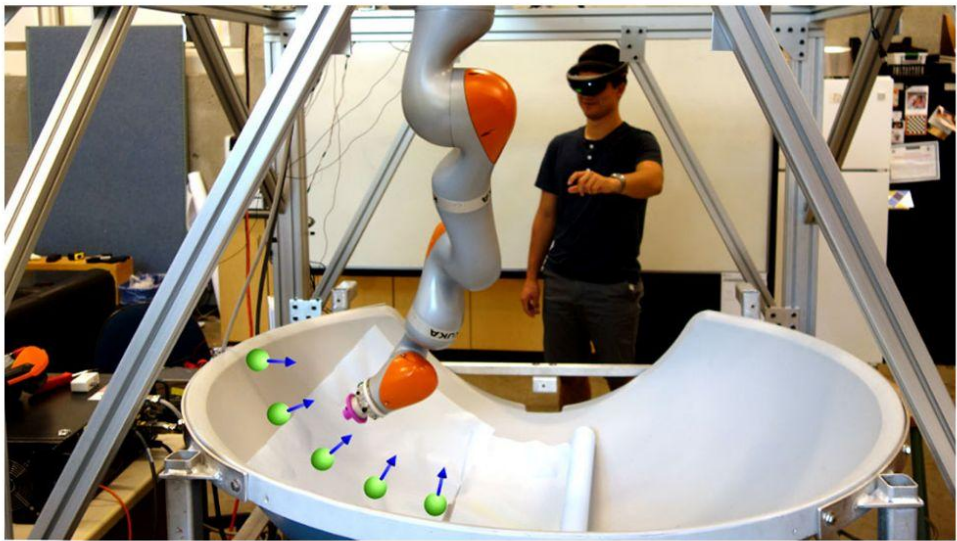
Literature Review

literature review 2:

Augmented-reality-assisted-human-robot-collaborative-manufacturing

<https://www.monash.edu/engineering/robotics/projects/augmented-reality-assisted-human-robot-collaborative-manufacturing>

particularly effective when applied to models that are highly detailed and complex in design, but relatively simple in the physical assembly process. In many cases, each component is unique—making traditional 2D drawings or manuals insufficient—while the overall construction logic remains repetitive, such as stacking bricks or assembling modules. This insight was reinforced by examples and demonstration projects showcased on the Fologram website, where AR is used to guide users through tasks involving complex geometry and unique part arrangements. Based on this understanding, we decided to design our own fabrication model with these same properties: geometrically rich and component-heavy, but logically simple to build, making it well-suited for human-robot collaboration assisted by AR.



Fologram examples

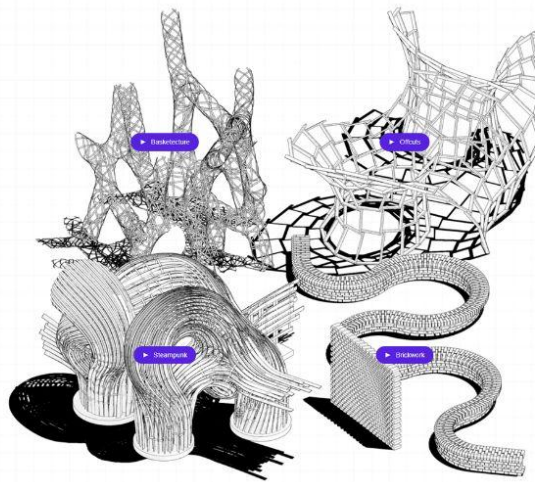
<https://fologram.com/examples>

PROJECT AND RESEARCH SHOWCASE

Hand-crafted architectural installations

Explore how designers and artists are taking advantage of skilled craft and creative improvisation to build beautiful structures in mixed reality. For ideas and inspiration from hundreds of projects built using Fologram, check out our project and research galleries in the Docs.

[VIEW ALL PROJECTS](#)



Part I

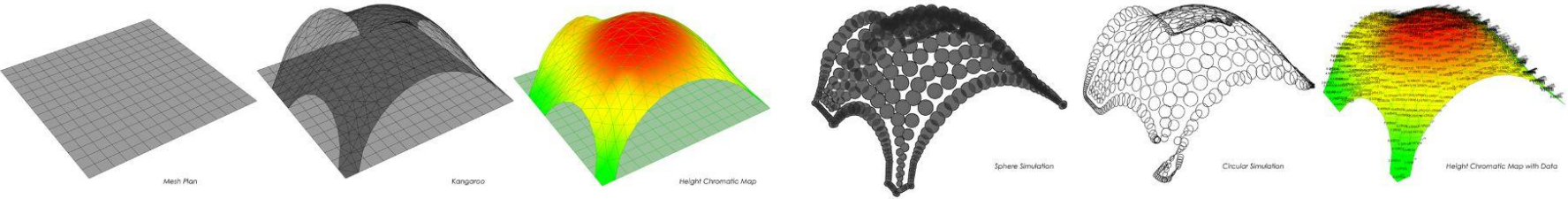
Morphological Exploration via Kangaroo Simulation

Using Kangaroo in Grasshopper, a series of parametric structural models were generated to explore structural behaviors under simulated physical forces. Four design iterations were tested, adjusting parameters to study how force affects form, leading to the selection of a stable and expressive geometry.

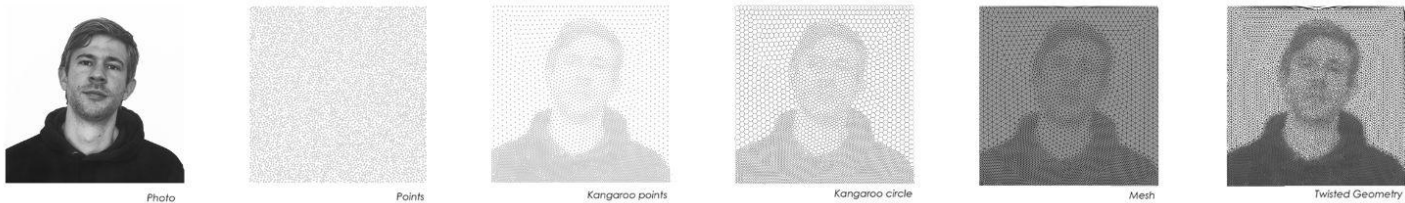


Morphological Study and Exploration with Kangaroo and Pufferfish

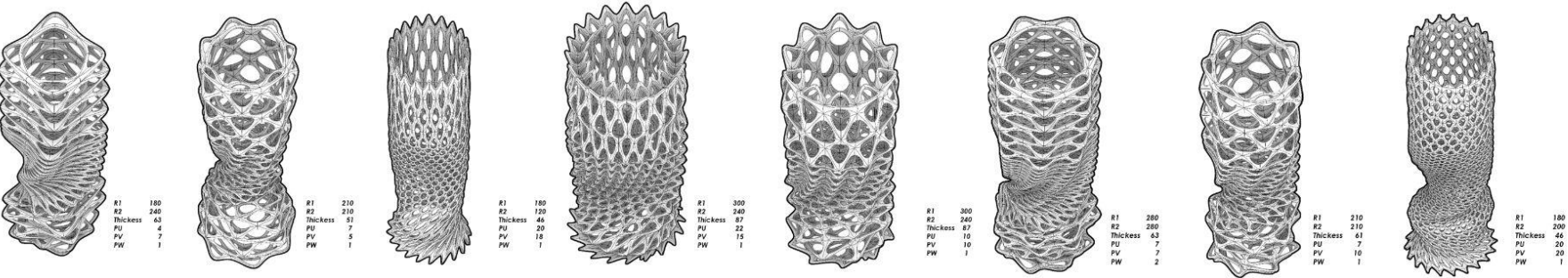
Tensile Structure Simulation with Kangaroo



Pattern Generation through Attraction and Collision Simulation with Kangaroo

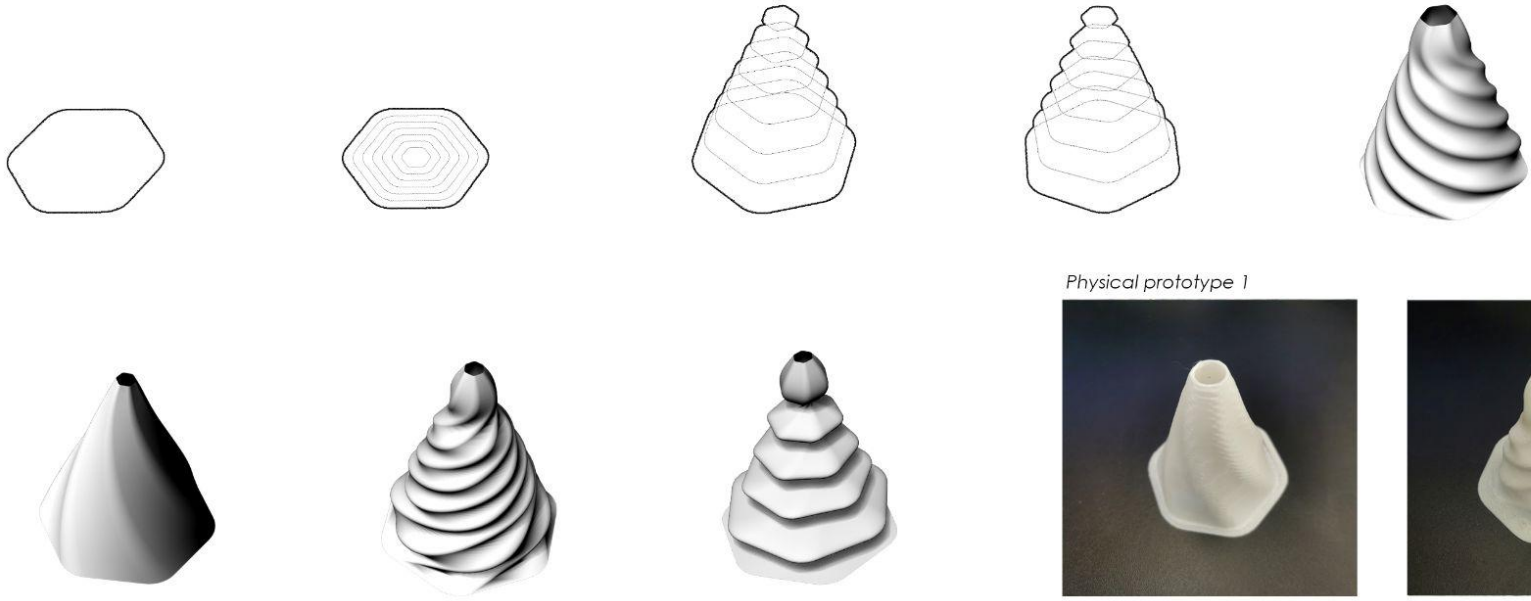


Simulating Ordered Layered Morphologies Using Pufferfish

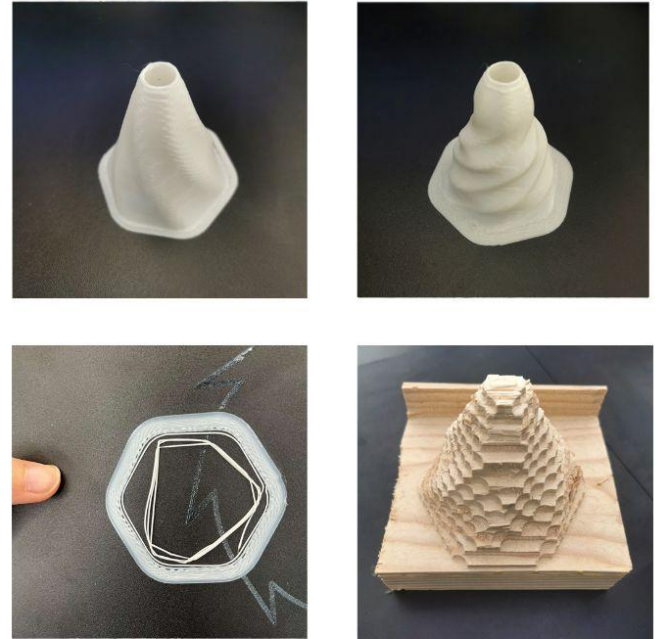


Model with Kangaroo

Prototype 1



Physical prototype 1

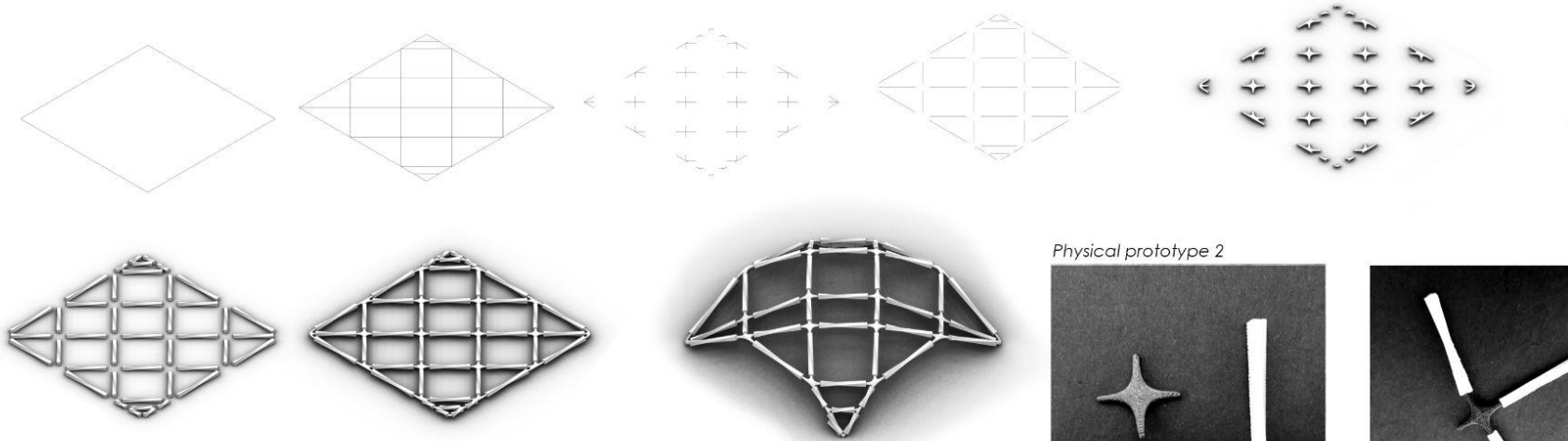


Summary:

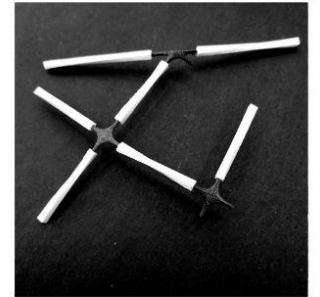
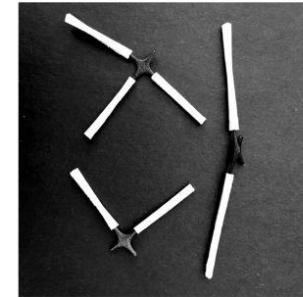
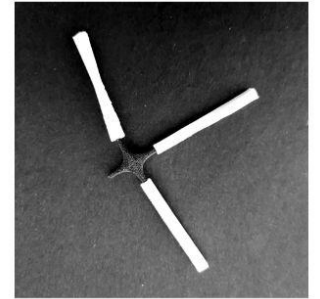
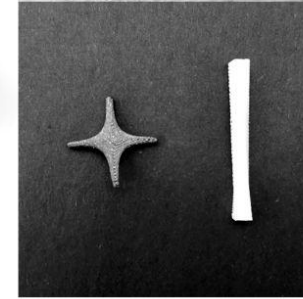
This initial phase explores the feasibility of using Grasshopper to generate G-code for controlling both 3D printing and robotic milling processes. Specifically, the workflow tests the successful translation of Grasshopper-generated geometry into executable G-code commands for additive manufacturing, and the use of robotic arm path planning for subtractive fabrication of timber materials. The objective is to verify the reliability of the G-code output, as well as to analyze the dimensional accuracy and spatial compatibility between milling paths and various drill bit geometries. These preliminary tests serve as a foundational validation of the digital-to-physical workflow, informing further development of fabrication logic and toolpath optimization.

Model with Kangaroo

Prototype 2



Physical prototype 2

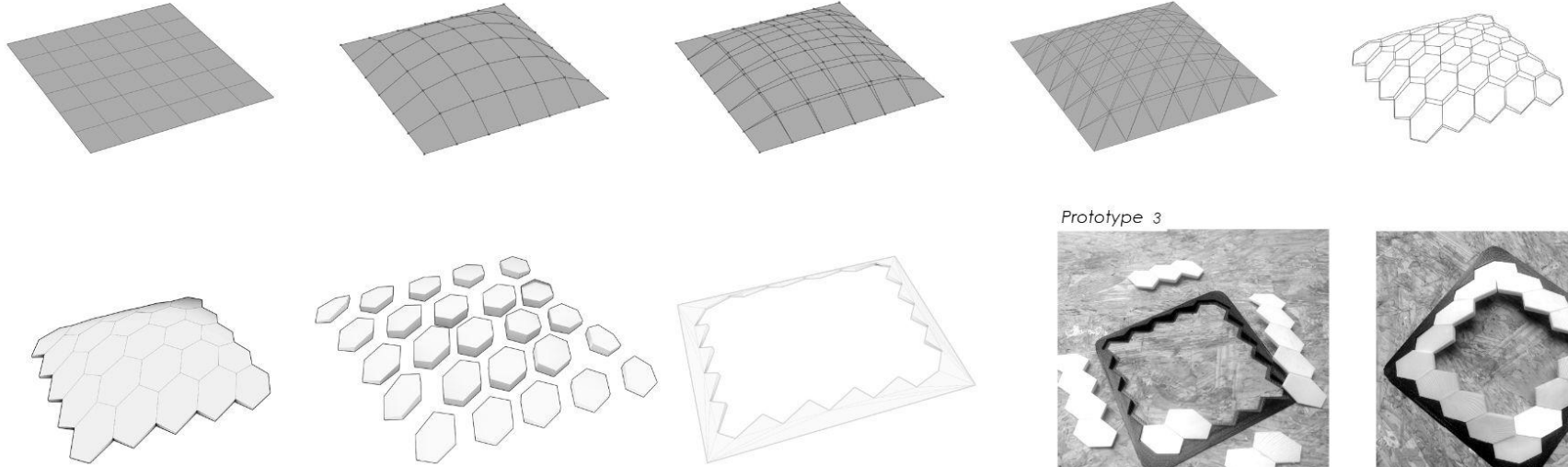


Summary

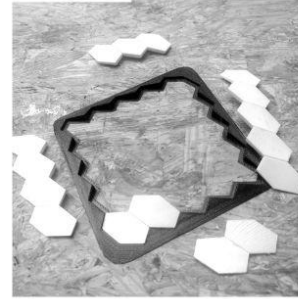
In this iteration, Kangaroo was used to generate a force-driven connection model, which was further developed with Pufferfish to create irregularly twisted beams and custom joints. While this approach enabled expressive and complex geometries, the 3D-printed joints exhibited insufficient structural strength and were unable to withstand the weight of the wooden components, leading to frequent breakage. Additionally, the insert-based connection system required extremely high assembly precision during the Fologram-guided process, which proved to be impractical on site. Due to these structural and assembly limitations, this scheme was ultimately abandoned.

Model with Kangaroo

Prototype 3

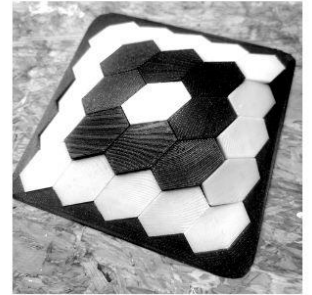
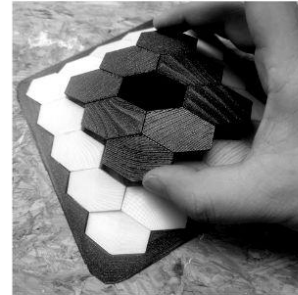


Prototype 3



Summary

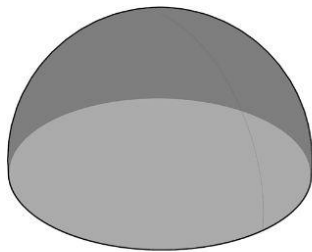
After generating a tensile structural model using Kangaroo, the form was segmented into multiple discrete components that could be assembled in a dome-like configuration. While this modular approach allowed for a flexible construction system, it also introduced significant structural instability during the early stages of assembly. Due to the lack of overall support before full completion, the components were prone to collapse. This issue became especially problematic during the Fologram-assisted assembly phase, where real-time control was compromised by the inherent fragility of the incomplete structure



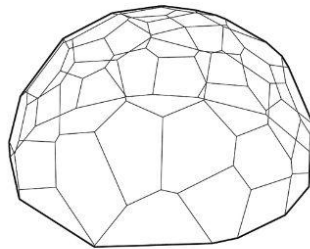
Final Model with Kangaroo



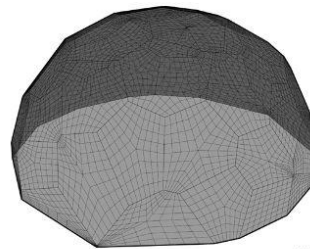
(i)



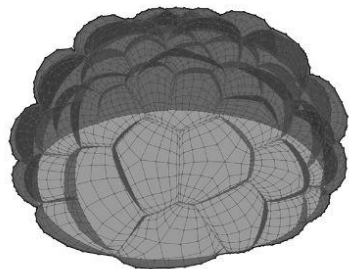
(ii)



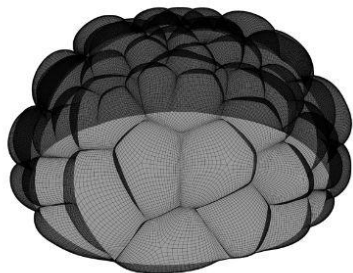
(iii)



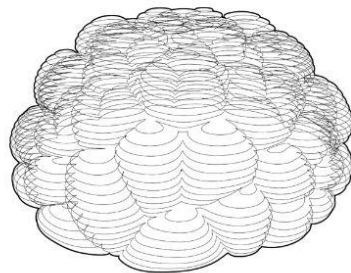
(iv)



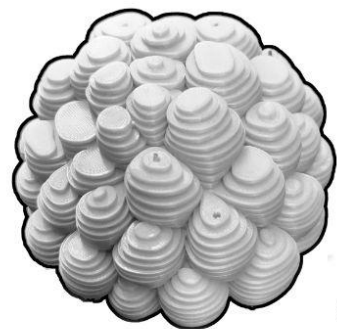
(v)



(vi)

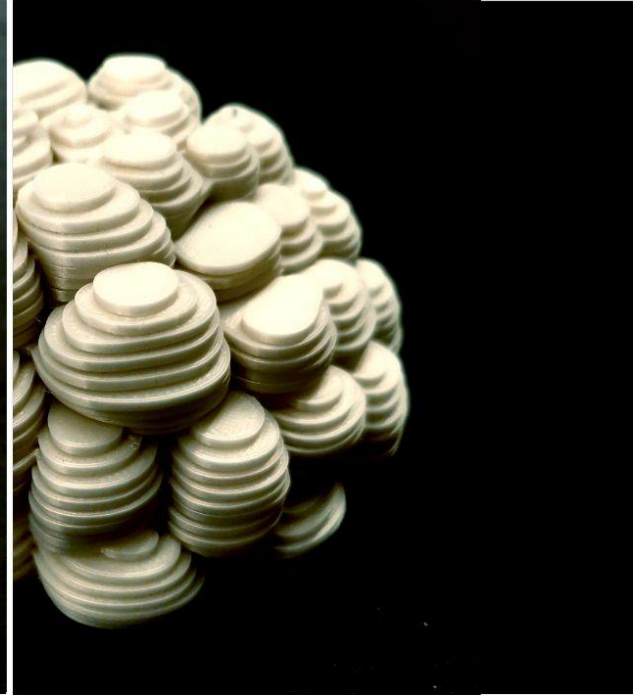


(vii)

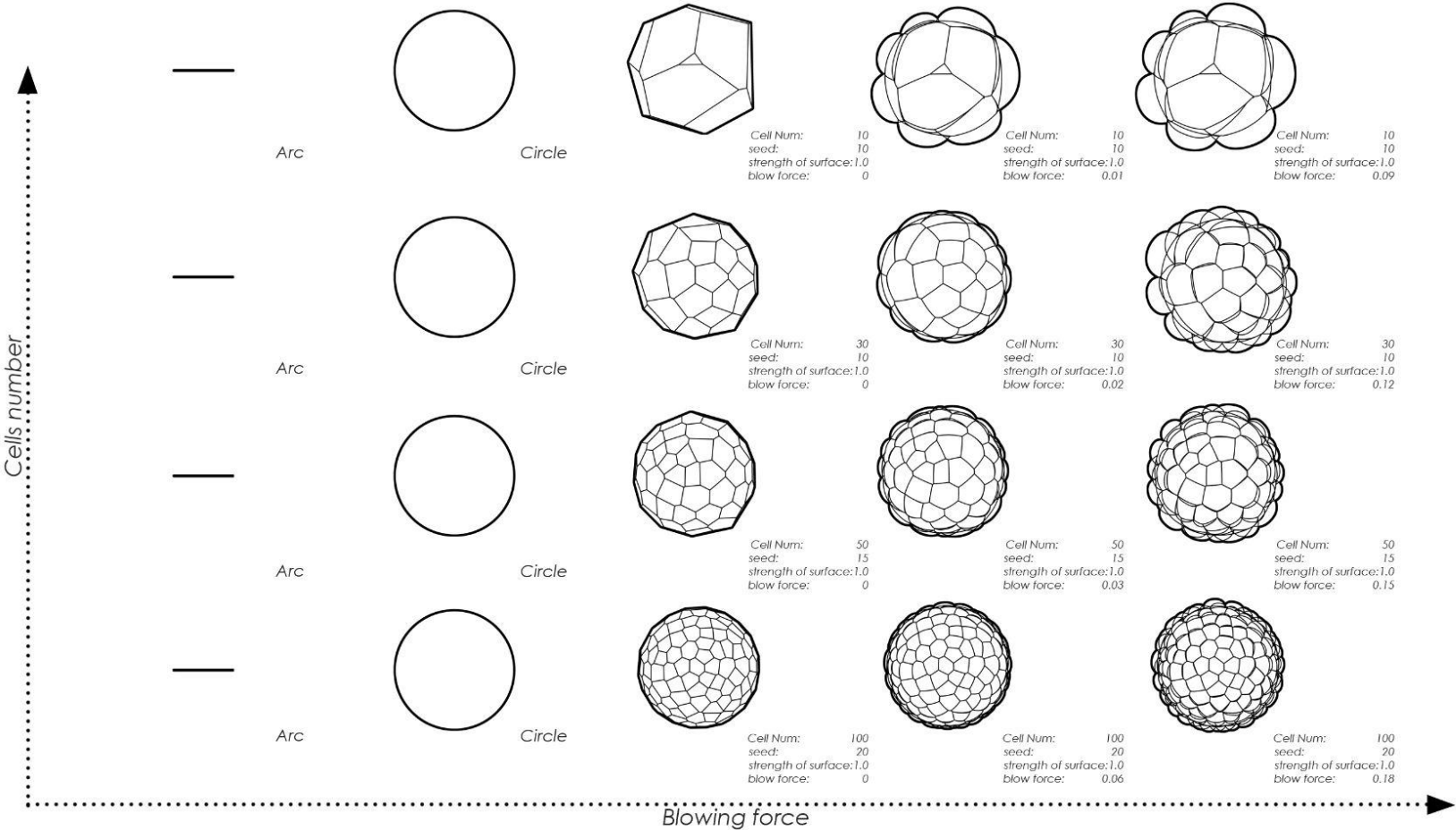


(viii)

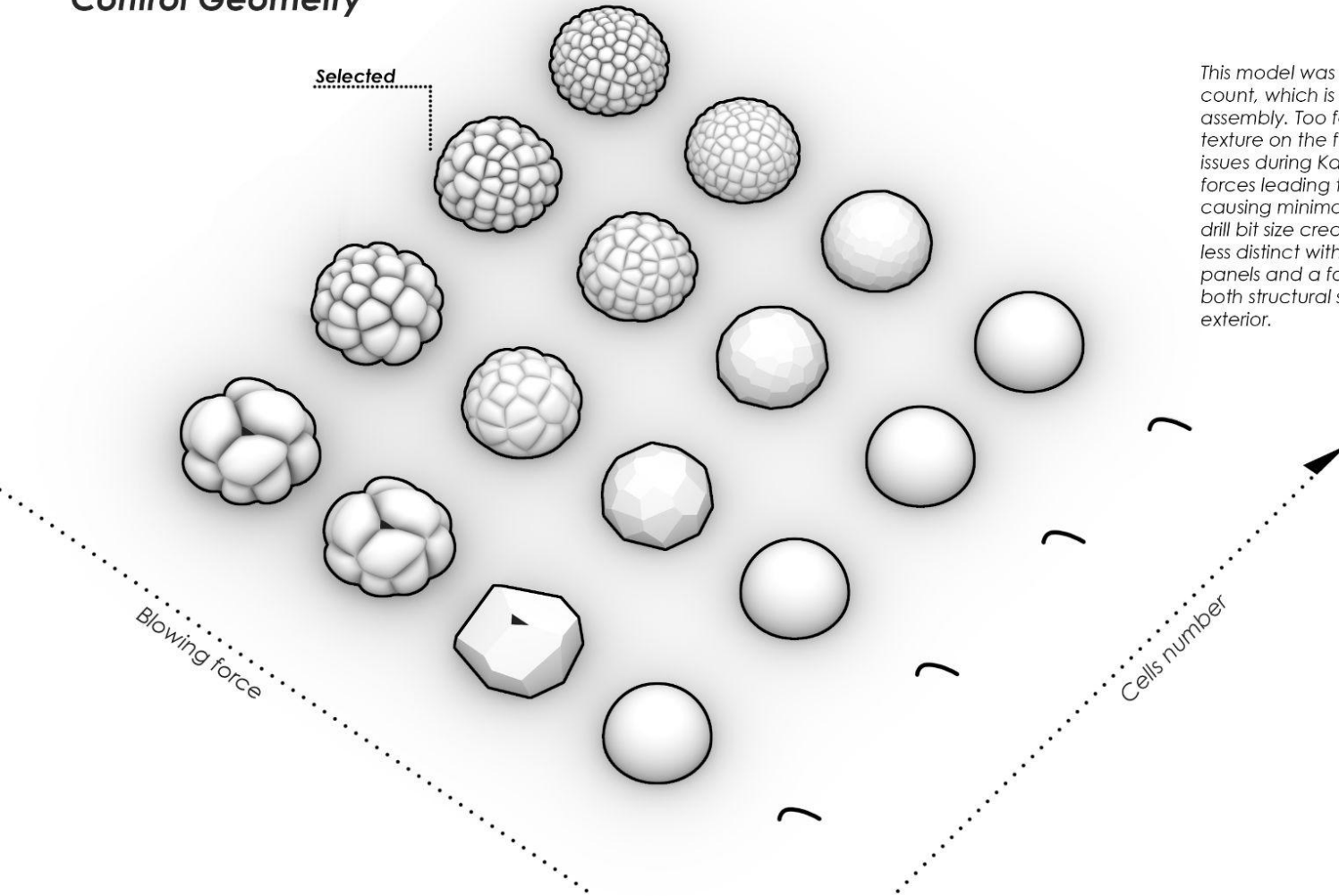
Final Model Photographs



Control Geometry

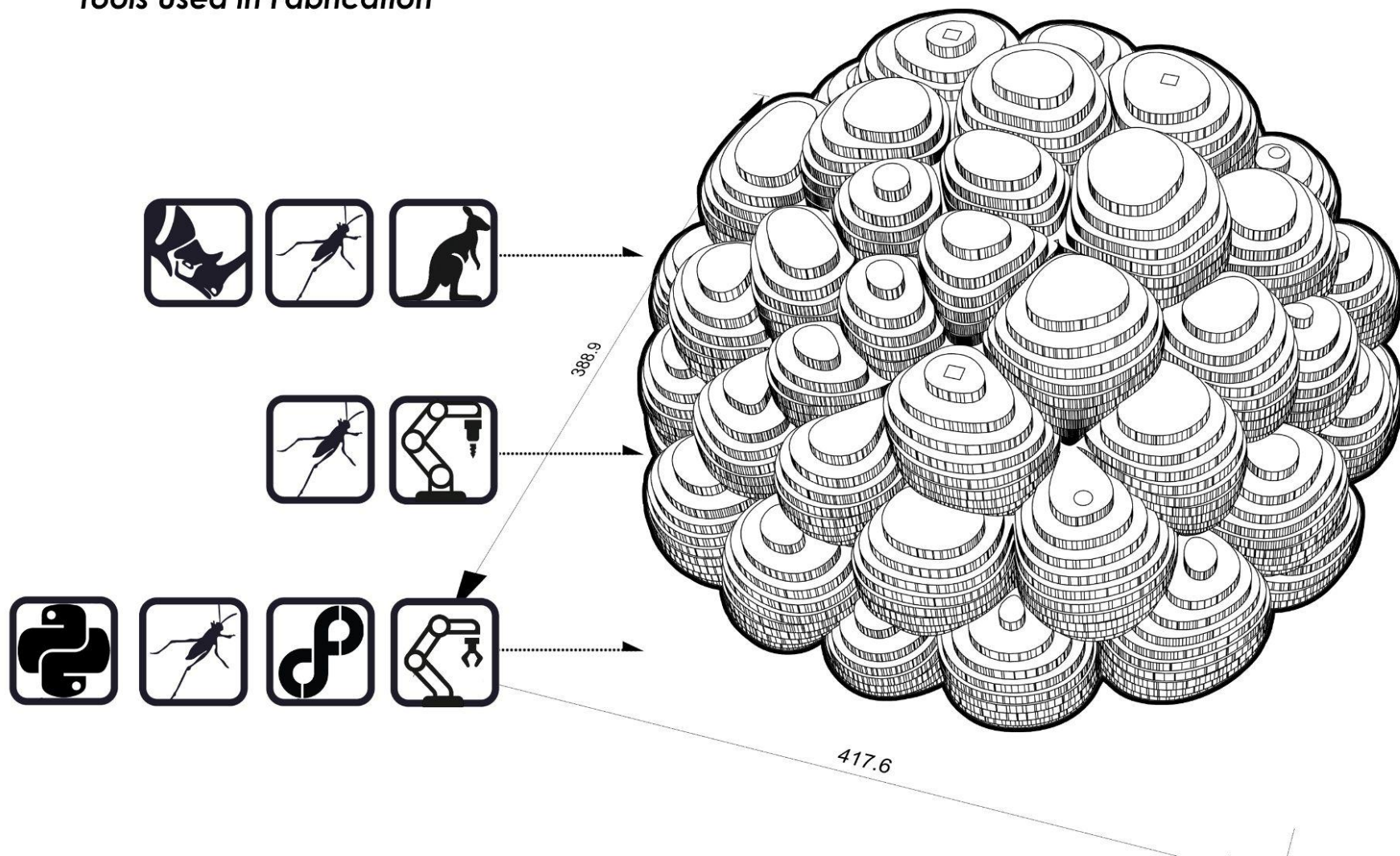


Control Geometry

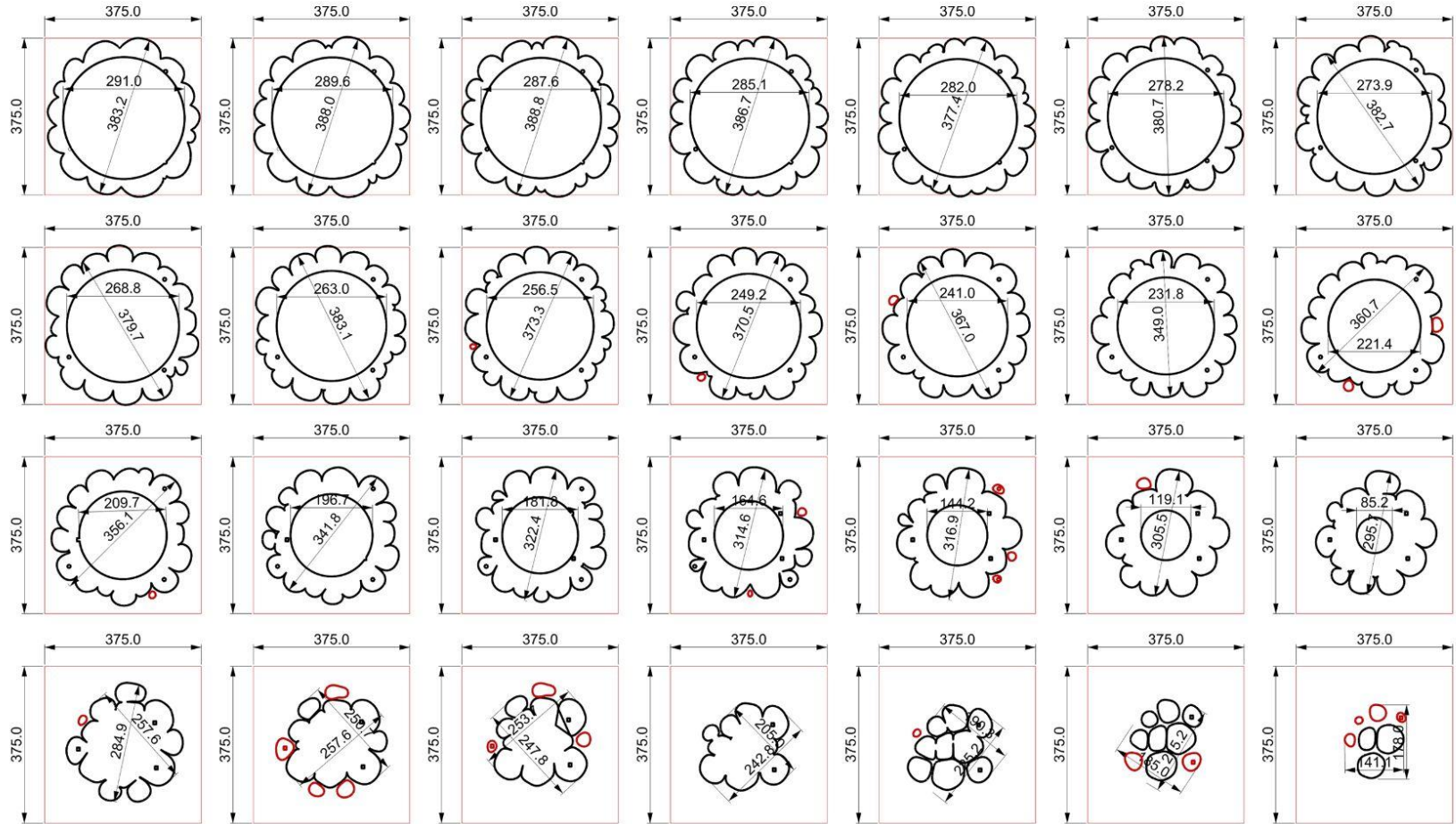


This model was selected due to its balanced panel count, which is optimal for robotic cutting and assembly. Too few panels would result in a lack of texture on the facade, while too many would cause issues during Kangaroo inflation, such as excessive forces leading to "explosion" or insufficient forces causing minimal protrusion. Additionally, the larger drill bit size creates a scale issue, making the facade less distinct with too many panels. A model with 50 panels and a force of 0.14 was chosen to ensure both structural stability and a clear, textured exterior.

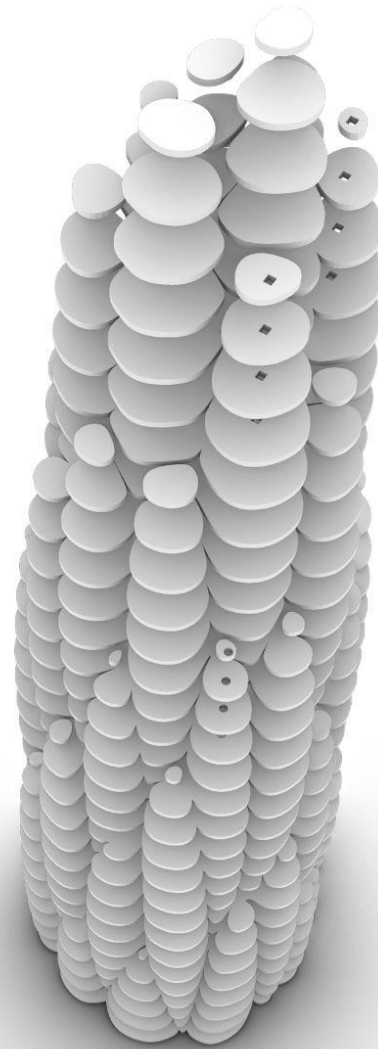
Tools Used in Fabrication



Isometric of Each Detail Type



Exploded Axon of Parts



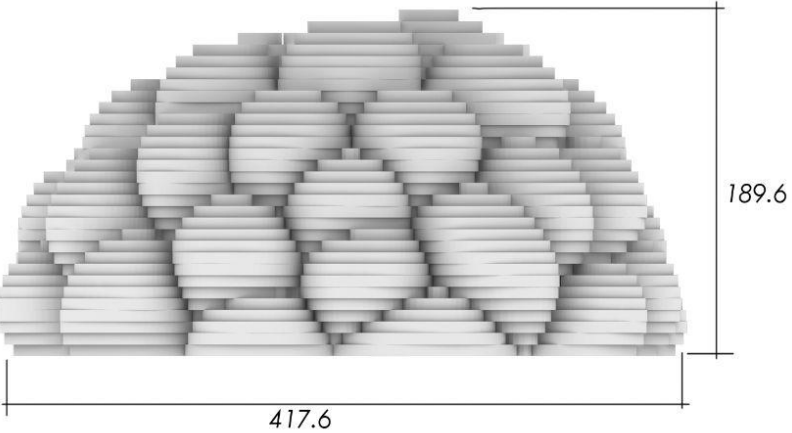
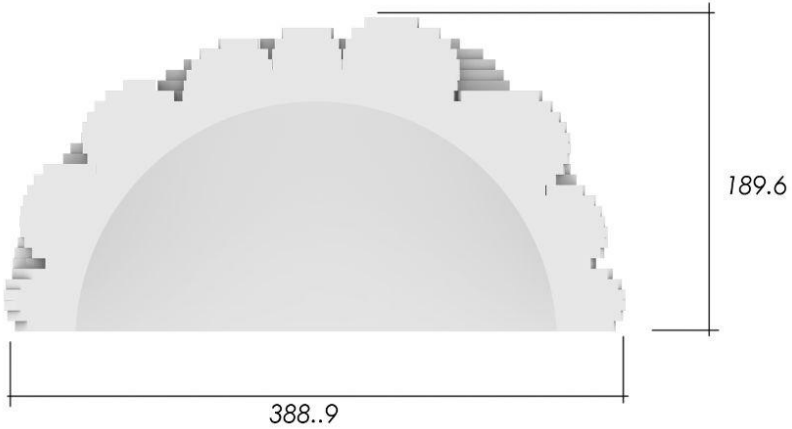
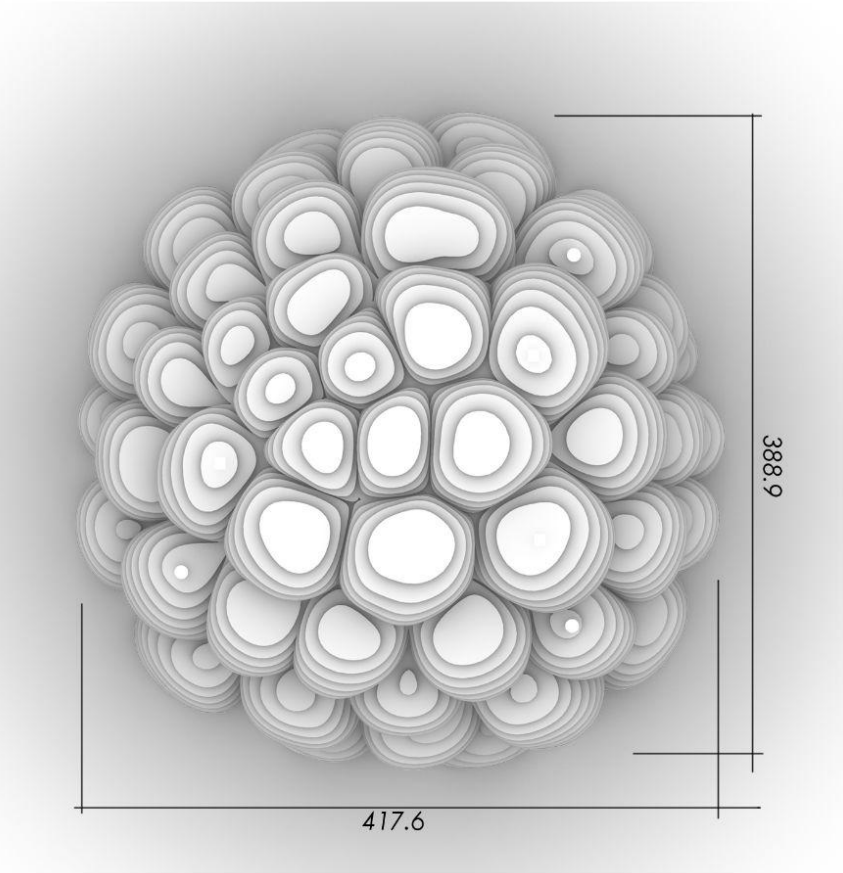
Part II

Robotic Fabrication Through Toolpath Scripting

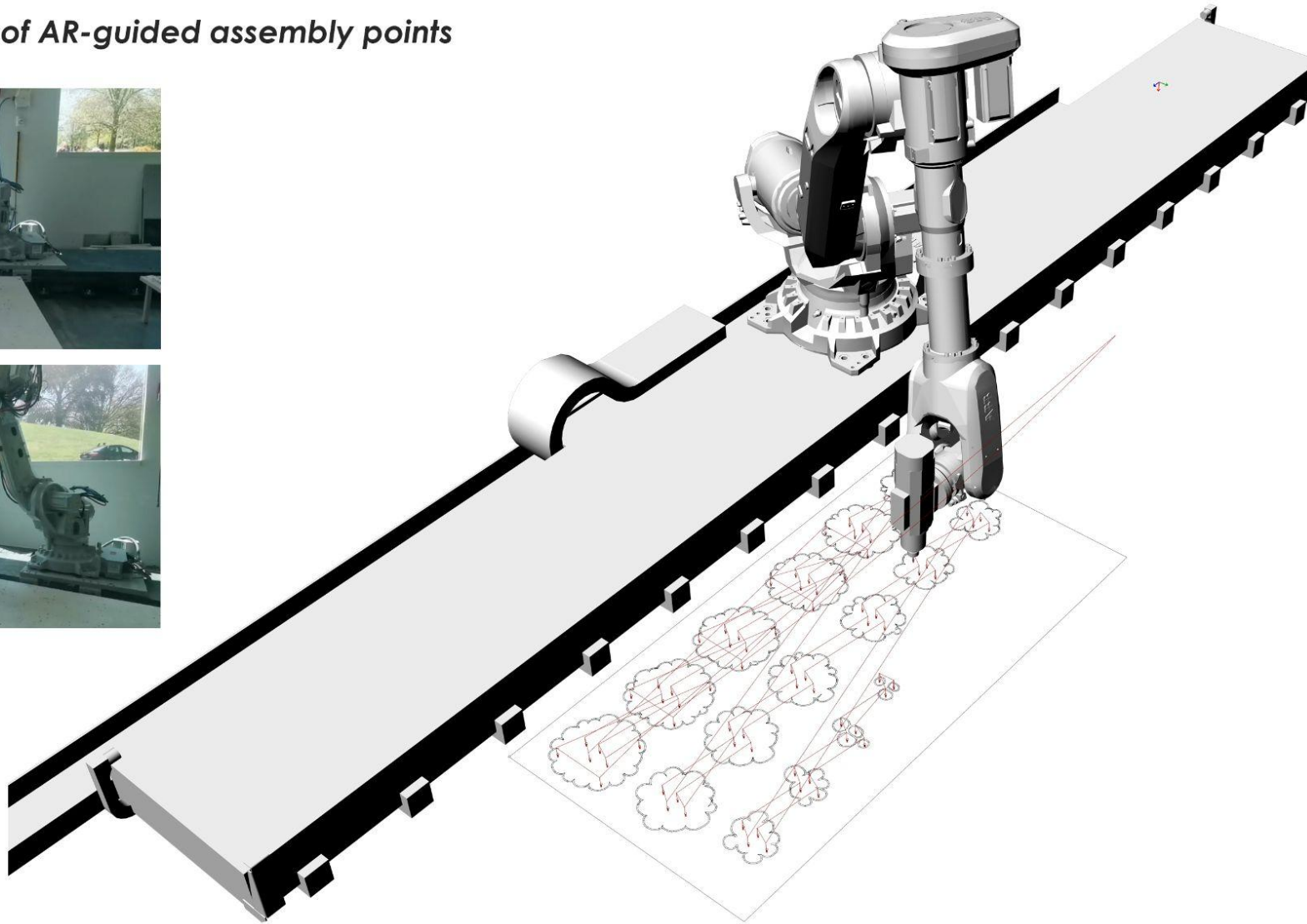
The finalized geometry was computationally sliced and converted into robotic toolpaths using Grasshopper in conjunction with specialized robotic fabrication plugins. The resulting scripts defined precise kinematic instructions for the robotic arm, enabling accurate layer-by-layer material processing in alignment with the digital design intent.



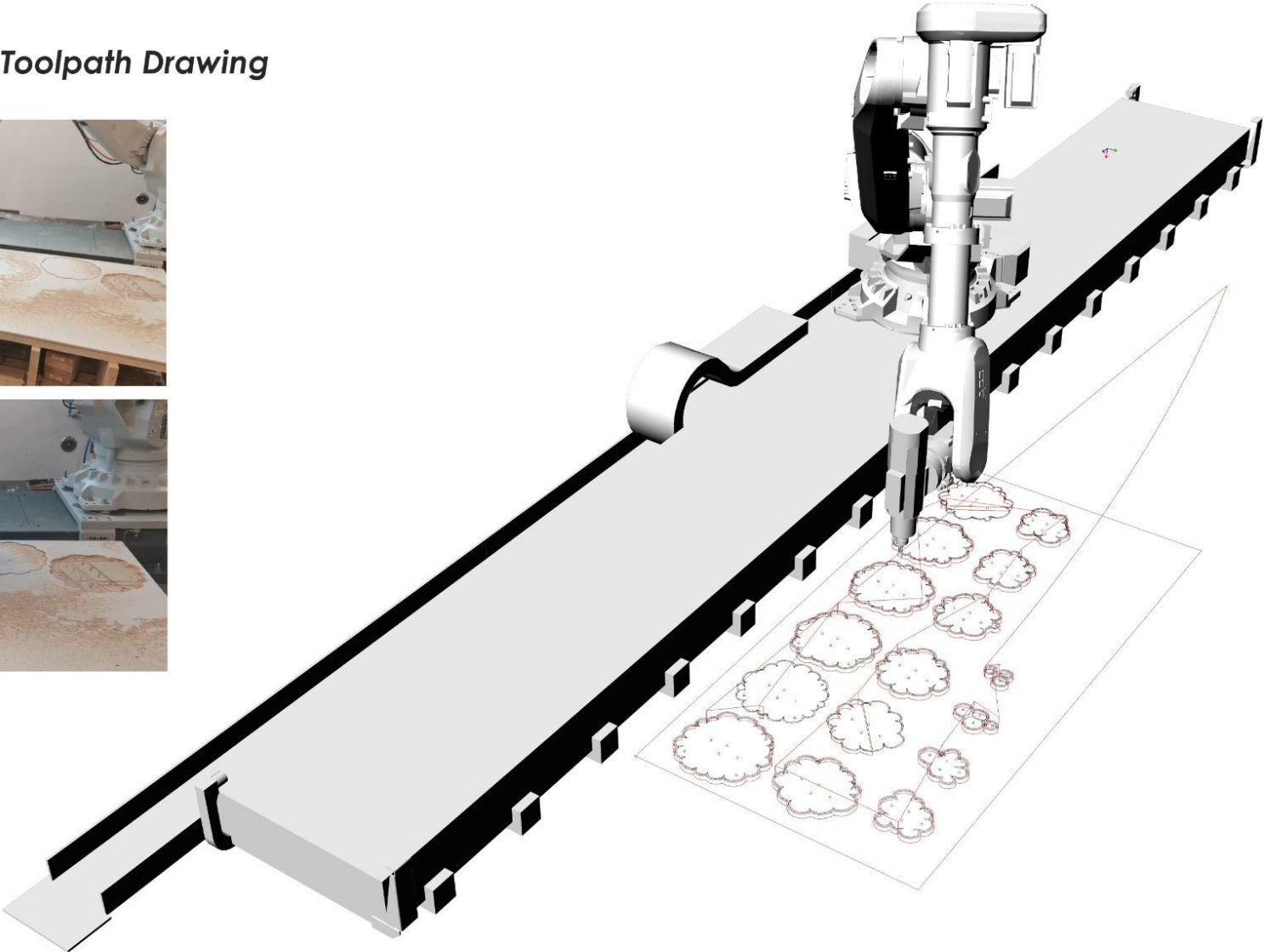
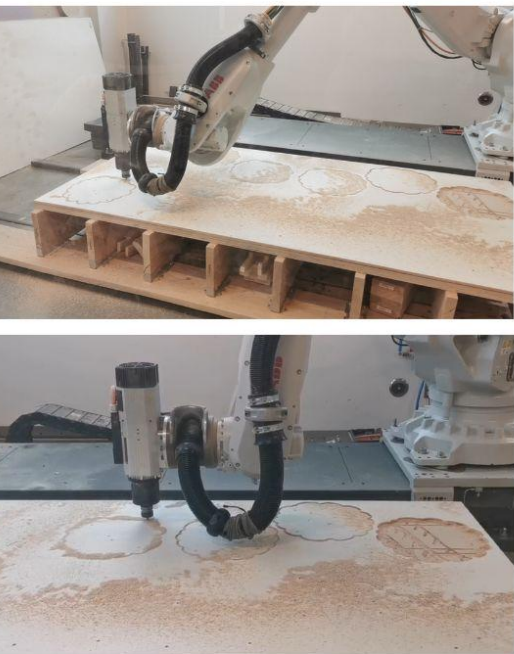
Fabrication Detail Drawing



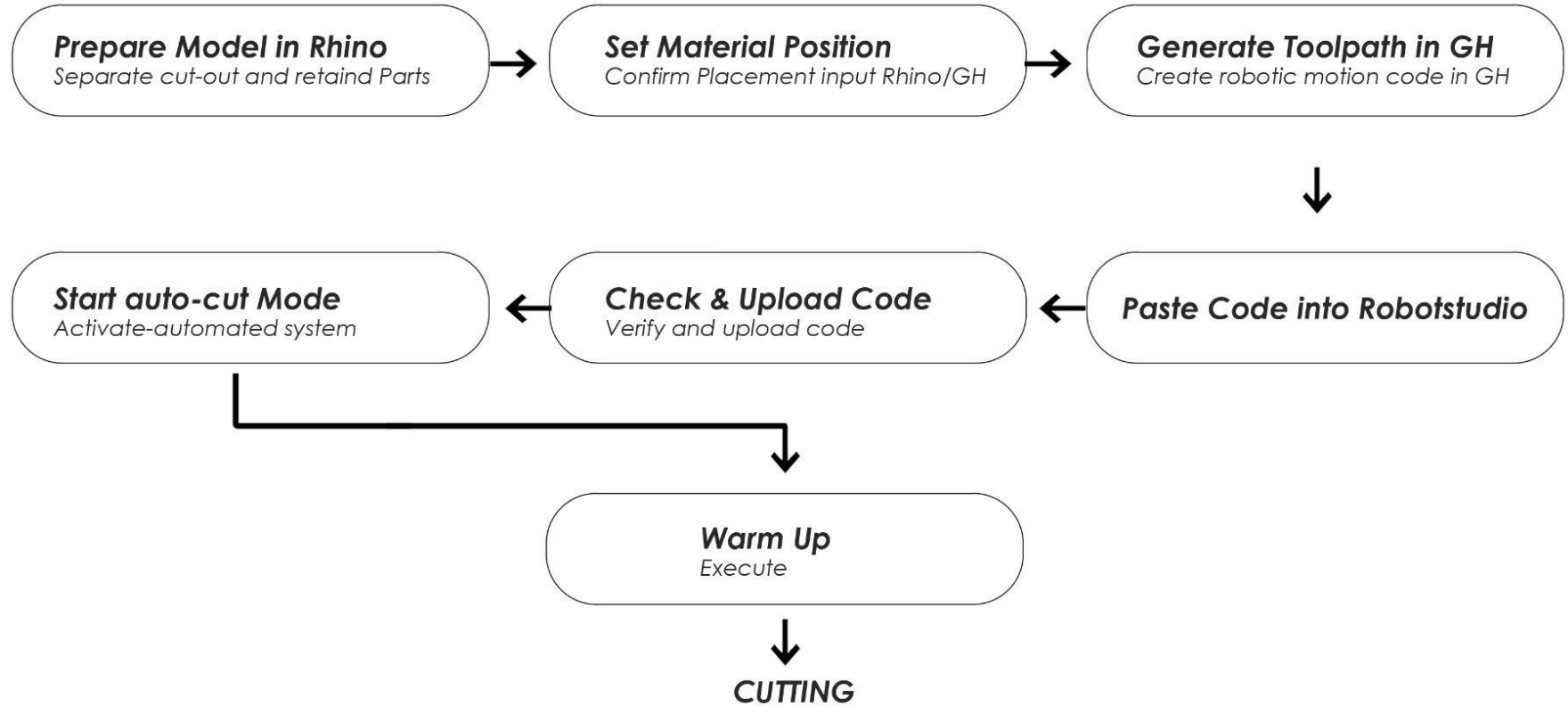
Fabrication of AR-guided assembly points



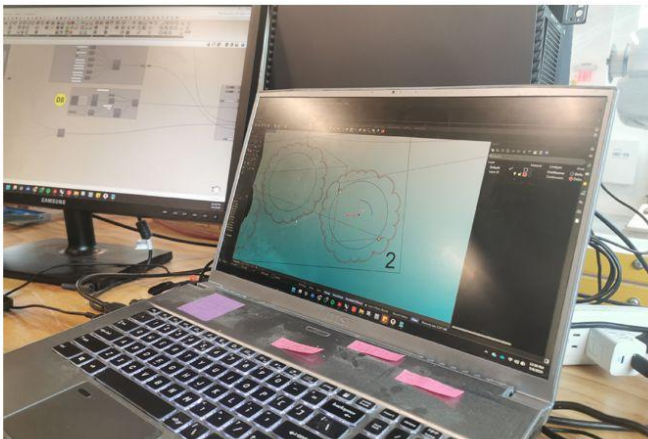
Fabrication Toolpath Drawing



Fabrication Methodology



Collage of Photos from Robotic Fabrication





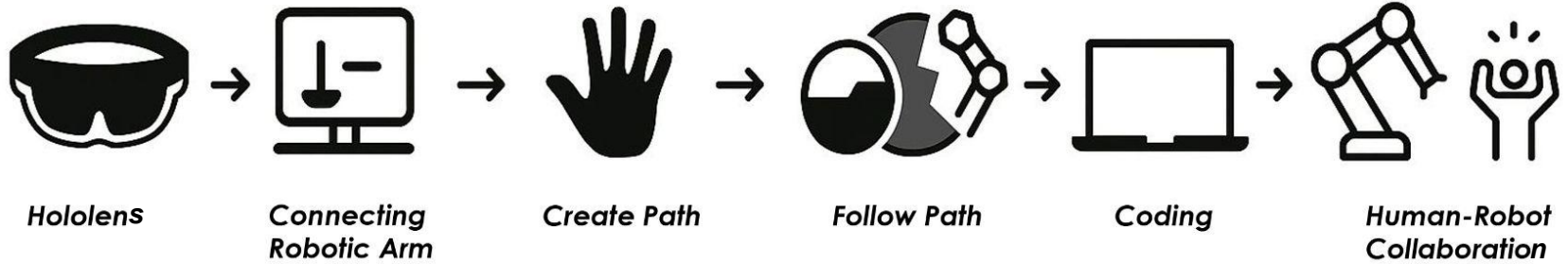
Part III

Fologram-Assisted Assembly via Human-Robot Interaction

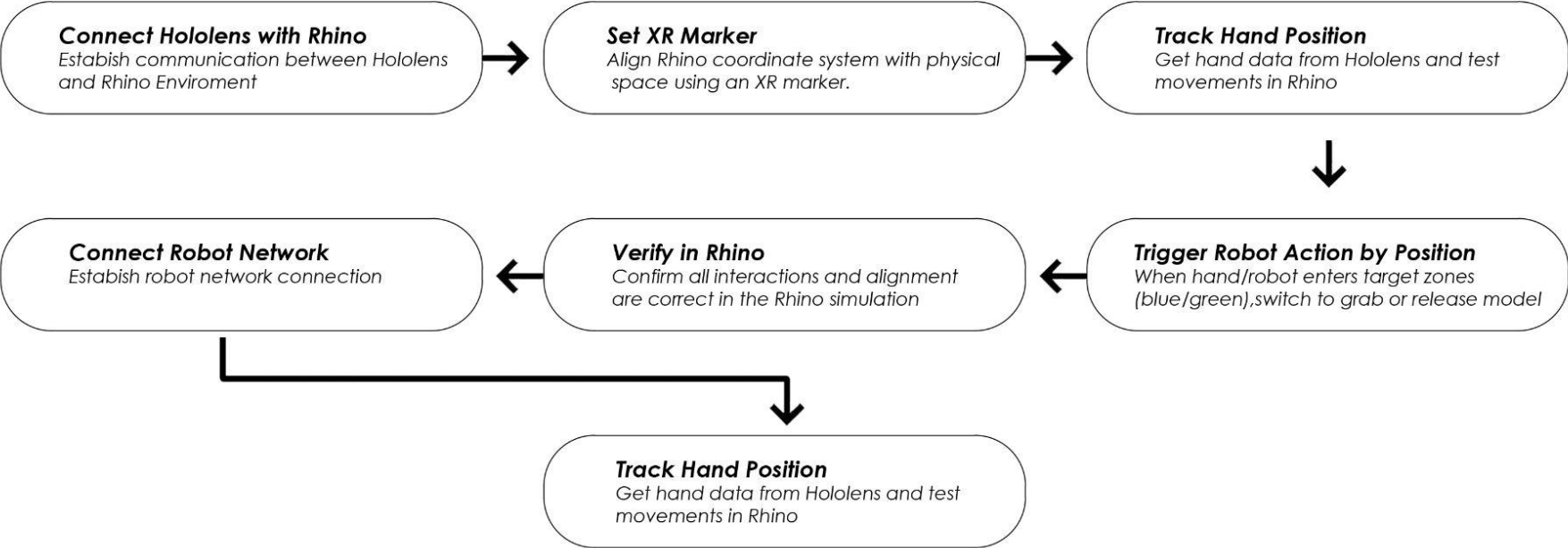
Augmented reality (AR) was employed through Fologram to facilitate an interactive, real-time robotic assembly process. Leveraging gesture recognition and spatial tracking, the system enabled direct human manipulation of the robotic arm, allowing for intuitive construction workflows without reliance on conventional programming interfaces.



Fologram Control Logic

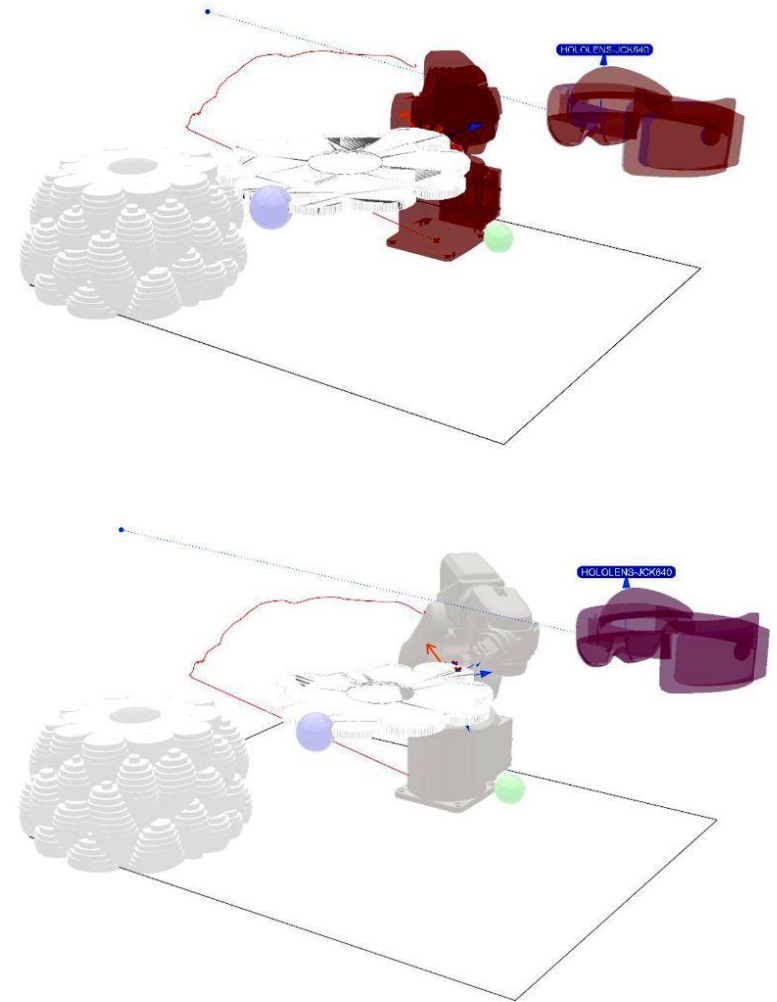
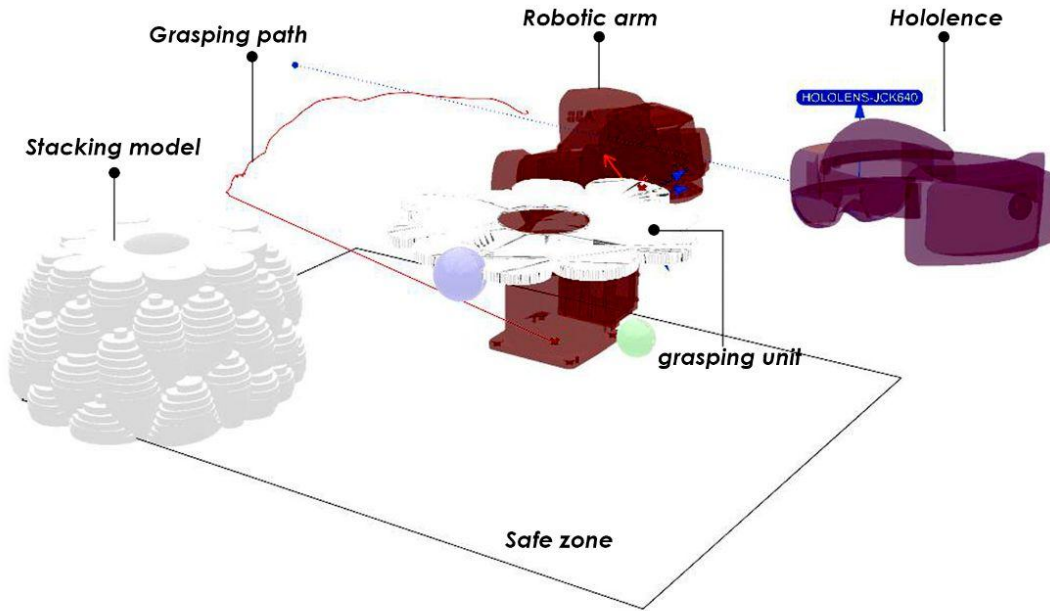


Fologram Methodology

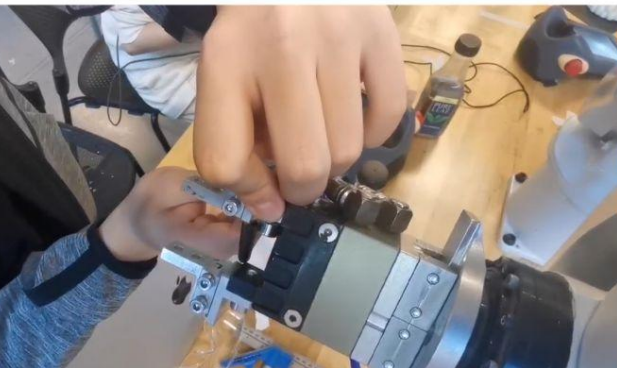


Toolpath for Fologram-Guided Assembly

"An AR-assisted system connects human input with a robotic arm for automated assembly. Using devices like the HoloLens, users manually define assembly paths in mixed reality, which are then captured and converted into motion commands. The robotic arm follows these paths to simulate and execute the assembly process autonomously, combining human intuition with robotic precision."



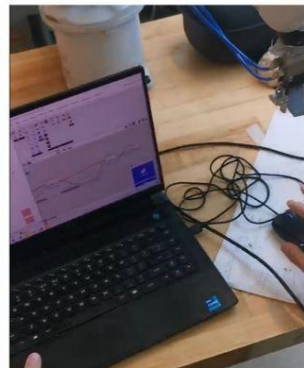
Collage of Photos from Fologram-Guided Assembly



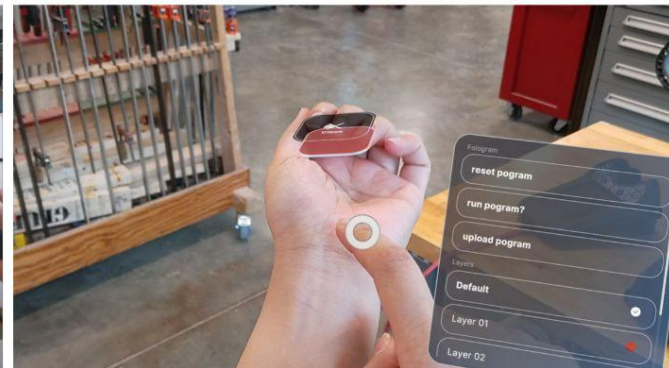
Arm Debugging



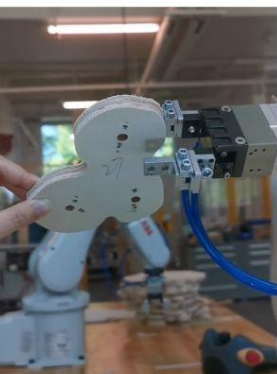
Coding Guidance



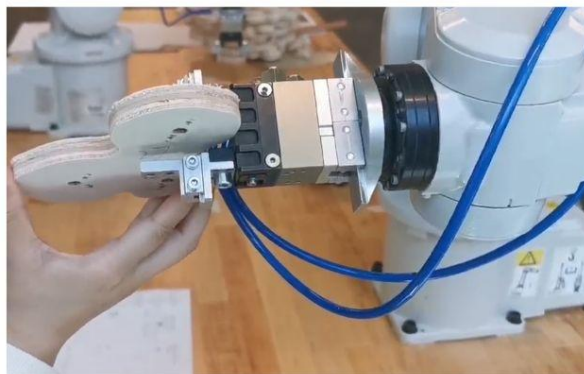
Program Development



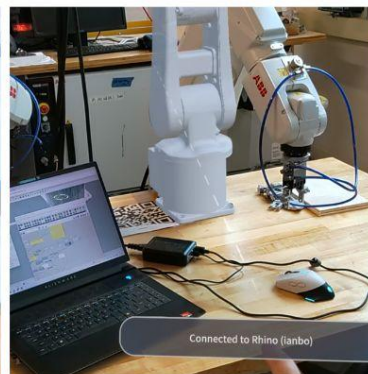
AR Control



Robotic Arm Grasping



Robotic Arm Grasping



Robotic Arm in HoloLens



Robotic Arm AR Simulation

Hard Points and What We Learned:

Mac Compatibility with ABB RobotStudio:

ABB RobotStudio is not Mac-friendly and cannot run natively on macOS. However, in this project, we overcame this limitation by using ABB's EGM (Externally Guided Motion) protocol, which transmits and receives data via WebSocket. This allowed us to connect to and control the robot arm from any device, including Macs, without relying on RobotStudio.

Stability of Rhino and Fologram:

Rhino and Fologram tend to crash when loading complex models or when moving hands during interactions. We resolved this issue by running the EGM client code independently—outside of the Python environment used by Rhino. This separation allowed Rhino to focus solely on data collection and simulation, significantly improving stability and preventing crashes.

Hand Visibility in HoloLens:

When using HoloLens to guide the robot arm, it is crucial that the user's hand remains visible to the device. In some cases, especially when the user tries to pull downward, the hand may become obscured—such as behind a wooden structure—causing tracking errors or collisions. Maintaining clear line-of-sight to the hand is essential for safe and accurate interaction.



Planned Developments

Transition from Open-Loop to Closed-Loop Control:

Currently, the method for placing wooden components involves moving the robot to an approximate position and releasing the piece. This approach lacks precision because it operates as an open-loop control system—we do not verify the robot's actual position before releasing. This limitation results in inaccurate placement.

In future development, we aim to implement closed-loop control by retrieving the robot's current position in real time. This would allow us to verify whether the robot has reached the intended target before placing the object, improving overall accuracy and reliability in the assembly process.

Assembly Limitations and Opportunities for Improvement:

While the robot was able to complete basic assembly tasks, it could not assemble every component with precision. Several constraints contributed to this:

Limited physical workspace restricted the robot's movement range.

HoloLens environmental tracking, which relies on visual data, still has limitations in accuracy and stability.

The small robot arm experienced unwanted flipping or rotation when navigating tight or awkward positions.

Potential improvements include:

Reducing the required movement range to minimize complex joint behavior.

Expanding the robot's physical workspace for smoother motion planning.

Introducing multi-arm collaboration, allowing multiple robot arms to share tasks and reach areas more efficiently.
